

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	AIML-C10
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: Voice-Based Multilingual Agricultural Advisory System using RAG and Answer Verification

Abstract

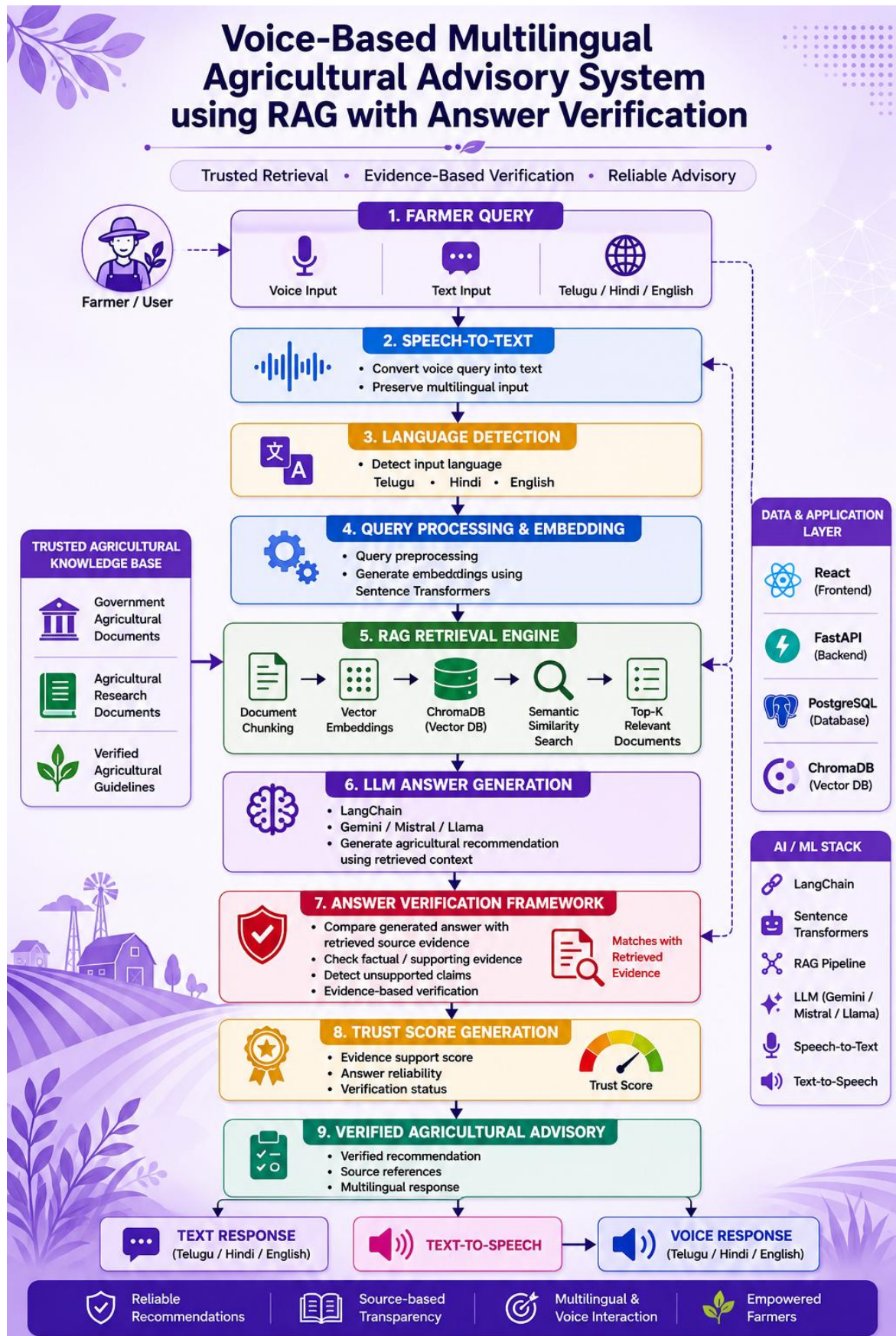
The increasing availability of digital agricultural information has created new opportunities for farmers to access crop-related guidance and recommendations. However, agricultural information is often fragmented across different sources, and existing AI-based chatbots may generate inaccurate or unsupported responses due to hallucinations. Such unreliable recommendations can affect decision-making in agriculture. To address these limitations, this project proposes a **Voice-Based Multilingual Agricultural Advisory System using Retrieval-Augmented Generation (RAG) with Answer Verification**, designed to provide reliable and transparent agricultural guidance to farmers through voice and text interaction.

The proposed system allows users to ask agricultural queries in **Telugu, Hindi, or English** using either voice or text. For voice-based interaction, the system first converts the farmer's speech into text and identifies the input language. The query is then processed by a **Retrieval-Augmented Generation (RAG) pipeline**, which retrieves relevant information from trusted agricultural documents. An LLM uses the retrieved information to generate an agricultural recommendation. Unlike conventional question-answering systems, the proposed system incorporates an **Answer Verification Module** that evaluates whether the generated response is supported by the retrieved source documents before presenting it to the user.

A key feature of the proposed system is the generation of a **Trust Score** for each response, providing an indication of the reliability of the generated recommendation. The system also presents the supporting source references to improve transparency and enable users to understand the basis of the recommendation. After verification, the final response can be delivered as text or converted into speech, enabling an accessible voice-based interaction for farmers. This approach aims to reduce unsupported AI-generated recommendations and improve the reliability of AI-assisted agricultural decision-making.

The system is implemented using **React** for the frontend, **FastAPI** for the backend, **PostgreSQL** for data management, and **ChromaDB** as the vector database. The AI pipeline incorporates **LangChain, Sentence Transformers, Retrieval-Augmented Generation, and an LLM such as Gemini, Mistral, or Llama**, along with Speech-to-Text and Text-to-Speech technologies for voice interaction. By integrating multilingual communication, voice-based interaction, trusted document retrieval, answer generation, answer verification, and trust-score generation, the proposed system provides a reliable, source-based, and transparent agricultural advisory platform for farmers and other agricultural users. The primary research contribution of the project is an **Answer Verification Framework for Agricultural RAG Systems** that evaluates whether generated agricultural recommendations are supported by trusted knowledge sources before they are presented to users.

Architecture Diagram:



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References (Category specific)

1. Leveraging Technology to Empower Millet Farmers: A Retrieval-Augmented Generation Approach with Large Language Models

Paper: [IEEE Xplore](<https://ieeexplore.ieee.org/document/10923869>)

Directly relevant to our RAG-based agricultural advisory system and use of LLMs for farmer assistance.

2. AgriBot: An Integrated Chatbot Platform for Precision Agriculture and Farmer Support using Deep Learning Techniques

Paper: [IEEE Xplore](<https://ieeexplore.ieee.org/document/10780432>)

Relevant to AI-based agricultural chatbots and farmer advisory services

3. FarmO'Cart: Multilingual Voice-Assisted Machine Learning Based Real-time Price Prediction to Enhance Agricultural Income

Paper: [IEEE Xplore](<https://ieeexplore.ieee.org/document/10193010>)

Relevant to our multilingual and voice-based interaction for agricultural users.

4. Agricultural Question Answering Dataset Creation and its Evaluation through BERT Based Experimental Analysis

Paper: [IEEE Xplore](<https://ieeexplore.ieee.org/document/10969374>)

Relevant to the agricultural question-answering component and evaluation of agricultural QA systems.

5. Certifying Generative AI: Retrieval-Augmented Generation Chatbots in High-Stakes Environments

Paper: [IEEE Xplore](<https://ieeexplore.ieee.org/document/10660589>)

6. A Retrieval-Augmented Generation Method Based on Intent Analysis and Ranking Fusion

Paper: [IEEE Xplore](<https://ieeexplore.ieee.org/document/10881002>)



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